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Deriving Relativistic Time Dilation from the Spatial Distribution of Mass: A Unified Origin of the Lorentz and Schwarzschild Metrics

Thomas Damon DeGerlia[a]

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ABSTRACT: The pace of time has meaning only relative to another pace of time. For two uniform spherical systems, this ratio is the compactness ratio $D_1/D_2 = (M_1R_2)/(M_2R_1)$, where $D = M/R$ is the DeGerlia compactness. This paper presents the relation and demonstrates numerical agreement with the standard general-relativistic calculation across six worked cases spanning classical-through-relativistic regimes. The static-mass case (a system compared with itself at its Schwarzschild radius) is the standard gravitational time-dilation calculation; the static-radius case (a linear, mass-only comparison) is the standard Lorentz time-dilation calculation.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

This paper compares the relative pace of time between two uniform spherical systems using only their masses and radii. The DeGerlia compactness $D = M/R$ [1] is the relevant property, and the ratio of two systems' compactness values gives the ratio of their paces of time.

When one system is compared with itself at its Schwarzschild radius, this ratio is the standard gravitational time-dilation calculation[2,3]. When two systems share a radius and differ only in mass, the ratio is the standard Lorentz time-dilation calculation[4]. The six worked cases in Section 6 and the appendix tables (Appendix Appendix A) demonstrate agreement with the standard general-relativistic calculation across the classical and relativistic regimes.

2 Inertial Relativity

Inertia is relativistic, meaning that without a comparator, any measurement of it would have no meaning. For example, something with a moment of inertia 1×10^{20} kg m^2 might be perceived as difficult to rotate, but not more difficult than something with 1×10^{30} kg m^2. However, the concept of relativity also says that how something is observed is influenced by the frame of observation. A system, in this context, is any collection of mass and spatial extent one chooses to define. It need not be spherical, uniform, high density, or even massive. A system may be a planet, a uniform sphere, a supermassive black hole, a disc galaxy, a cube, a cubic sample of a cumulus cloud, or a system consisting of a hydrogen proton and a single electron.

2.1 Rotational Inertia

Moment of inertia dictates how an applied torque translates into angular acceleration. It is defined as:

$$I = \sum_i m_i r_i^2 \tag{1}$$

where r_i is the perpendicular distance from each mass element m_i to the rotation axis. Just as mass governs how a system responds to linear force ($F = ma$), moment of inertia governs how it responds to torque ($\tau = I\alpha$).

2.2 Linear Inertia

Linear inertia is mass:

$$M = \sum_i m_i \tag{2}$$

where m_i are the constituent mass elements. Linear inertia defines how an applied force translates into translational acceleration. It is quantified entirely by mass ($F = ma$). It is dimensional, operating along a specific spatial vector, but unlike moment of inertia, it is isotropic: mass offers the same resistance to changes in motion regardless of the direction of the vector.

[a] DeGerlia Expert Consulting, 3000 Lawrence Street, Denver, CO, United States of America. E-mail: tom.degerlia@tomdegerlia.com

2.3 Isometric Scaling

Isometric scaling is the uniform scaling of a system in all spatial dimensions while maintaining constant density. Under isometric scaling, every length in the system scales by the same factor k:

$$\ell' = k\ell \tag{3}$$

where ℓ is the characteristic length of the system and k is the linear scale factor. Isometric scaling of a system by a factor of k results in the following changes to mass, radius, and moment of inertia:

$$M' = k^3 M \tag{4}$$

where M is the mass and k is the linear scale factor.

$$R' = kR \tag{5}$$

where R is the radius.

$$I' = k^5 I \tag{6}$$

where I is the moment of inertia and k is the linear scale factor.

2.4 Inertia Abstracts Mass and Radius

Because every change in length necessarily changes the system's mass, Equations 3–5 hold only for strict isometric scaling; Equation 6, however, abstracts mass and radius into moment of inertia. If two systems share moment of inertia, then an identical torque applied to each system will produce identical changes in motion, regardless of the system geometry; independent of mass and radius individually. The moment of inertia alone dictates how a system interacts with time. From this, we can conclude, for all systems that share that same density, regardless of geometry the following formula is universal and is not bound in any way to isometric scaling. The derivations that follow demonstrate that General and Special Relativity emerge from this principle.

2.5 Universality

For any two uniform spherical systems, the characteristic length scale factor that determines time dilation is the fifth root of the inverse ratio of their moments of inertia, multiplied by the ratio of their densities to the 3/10 power (Equation 16). Stated more cleanly, it is the ratio of the respective DeGerlia compactness D values for the compared systems (Equation 10).

3 Inertial Density P

Inertial density P[1] is the moment of inertia of a system divided by its volume:

$$P = I/V \tag{7}$$

Since linear inertia is mass, this reduces directly to mass density $\rho = M/V$ in the linear case; inertial density is the rotational generalization of mass density, a quantity already familiar from linear mechanics. Or more accurately, mass density is simply the linear interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \tag{8}$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

3.1 DeGerlia Compactness D

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

3.2 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)[3] and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26}\ \text{kg/m} \tag{9}$$

Note this value is exact and accurate to the limit of precision imposed by G, which is accurate to six significant figures with a margin of error of $15/10^6$[5].

4 The Space-Time Equivalence

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k_d = \frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1}\right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{10}$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i. The three forms in Equation 10 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1R_2/(M_2R_1)$ is the fully expanded form.

The pace of time has meaning only relative to another pace of time. The compactness ratio k_d governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation for a single system takes the Schwarzschild form $\sqrt{1-k_s}$, where $k_s = D/D_{crit} = r_s/R$. The compactness ratio $k_d = D_1/D_2$ and the Schwarzschild form k_s are the same scale factor k evaluated by two routes; the subscript

* This identity has been previously referred to as the "space-time equivalence"[6].

denotes which route produced it.

Under isometric scaling (constant density), the isometric equivalence holds:

$$k_i = \left(\frac{I_1}{I_2}\right)^{1/5} \tag{11}$$

where k_i is the linear scale factor derived from the inertia ratio. Under isometric conditions the two systems share the same density, and the fifth root of the inertia ratio recovers the linear scale factor exactly. Subsequent references to this specific inertia-derived form use k_i.

In the linear context, inertia is M and the linear inertia version of the isometric equivalence applies (see Section 6). Special Relativity describes linear inertia:

$$k_m = \left(\frac{M_1}{M_2}\right)^{1/3} \tag{12}$$

where M_1 and M_2 are the masses of the two systems and k_m is the linear scale factor derived from the mass ratio. Subsequent references to this specific mass-derived form use k_m.

Via the Schwarzschild radius, the same scale factor is:

$$k_s = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{13}$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. k_s is the Schwarzschild-route name for k that appears in the standard $\sqrt{1-k}$ for GTD.

4.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{14}$$

where $d\tau/dt$ is the ratio of proper time to coordinate time and k is the compactness ratio for the case being computed. The specific forms are:

- $k_d = D_1/D_2$ (Equation 10) — via inertial density ratio
- $k_s = D/D_{crit} = r_s/R$ (Equation 13) — via Schwarzschild radius
- $k_i = (I_1/I_2)^{1/5}$ (Equation 11) — via moment of inertia ratio (constant density)
- $k_m = (M_1/M_2)^{1/3}$ (Equation 12) — via mass ratio (static radius)

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4.2 STE Example

Setup. A uniform sphere of mass 1 kg and radius 1 m (System 2) is scaled isometrically by a factor of two, producing a sphere of mass $2^3 = 8$ kg and radius 2 m (System 1). The linear scale factor between them is $k = R_1/R_2 = 2$ by construction.

Change in inertia. The moments of inertia are

$$I_1 = \tfrac{2}{5}(8)(2)^2 = 12.8\,\mathrm{kg\,m^2}, \qquad I_2 = \tfrac{2}{5}(1)(1)^2 = 0.4\,\mathrm{kg\,m^2},$$

so $I_1/I_2 = 32 = 2^5 = k^5$ — the moment of inertia changes by the fifth power of the linear scale factor.

Gravitational time dilation. Each system's Schwarzschild radius is $r_s = 2GM/c^2$, giving $r_{s,1} = 1.18819 \times 10^{-26}$ m and $r_{s,2} = 1.48523 \times 10^{-27}$ m. The Schwarzschild gravitational time dilation $gtd = \sqrt{1 - r_s/R}$ is then

$$gtd_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = 1 - 2.97046 \times 10^{-27},$$

$$gtd_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = 1 - 7.42616 \times 10^{-28},$$

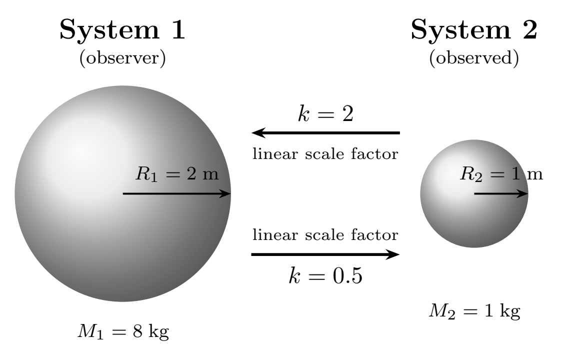
so the relative time dilation between the two systems is

$$\frac{gtd_1}{gtd_2} = 1 - 2.22785 \times 10^{-27}.$$

Compactness ratio. The compactness ratio follows from the inertial densities $D = M/R$ ($D_1 = 4$ kg/m, $D_2 = 1$ kg/m):

$$k_d = \frac{D_1}{D_2} = \frac{4}{1} = 4 = k^2, \qquad \sqrt{k_d} = 2 = k.$$

This matches the STE Example column of Appendix Appendix A: the gravitational time dilation ratio is $1 - 2.22785 \times 10^{-27}$ and $\sqrt{k_d} = 2$.



Two uniform spheres related by isometric scaling, scale factor $k = 2$: System 1 ($M_1 = 8$ kg, $R_1 = 2$ m) and System 2 ($M_2 = 1$ kg, $R_2 = 1$ m). The moment of inertia changes by $2^5 = 32$; the relative time dilation $1 - 2.22785 \times 10^{-27}$ is obtained equivalently from the Schwarzschild and inertial-density formulations.

4.3 Density-Explicit Form

The STE (Equation 10) can be expressed equivalently in terms of density. With $I = \frac{2}{5}MR^2$ and $M = \frac{4}{3}\pi R^3 \rho$, so $I \propto \rho R^5$. Taking the fifth root of the inertia ratio (k_i from Equation 11) gives:

$$\left(\frac{I_1}{I_2}\right)^{1/5} = \left(\frac{\rho_1}{\rho_2}\right)^{1/5} \cdot \frac{R_1}{R_2} \tag{15}$$

Substituting Equation 15 into the STE (Equation 10) and resolving the radius ratio in terms of density yields the universal STE in (I,ρ) coordinates:

$$\sqrt{k_d} = \left(\frac{I_1}{I_2}\right)^{1/5} \cdot \left(\frac{\rho_1}{\rho_2}\right)^{3/10} \tag{16}$$

The 3/10 exponent ($= 1/2 - 1/5$) on the density ratio arises from the difference between the density dependence in $\sqrt{k_d}$ (1/2 power) and the inertia ratio (Equation 11, 1/5 power). Under constant density this factor is 1 and k_i (Equation 11) gives the compactness ratio directly. Equation 16 is the universal STE; k_i is its constant-density case. The pair comparison tables (Appendix Appendix A) demonstrate the universal form's necessity in the non-isometric cases, where the density correction is not 1.

5 General and Special Relativity

Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}} \tag{17}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and R is the radius.

Lorentz time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \tag{18}$$

where v is the velocity of the moving frame and c is the speed of light.

These are the same form as Equation 14, where $k = r_s/R$ in the gravitational case and $k = v^2/c^2$ in the velocity case. The two expressions are equivalent: the escape velocity at radius R is $v^2 = 2GM/R$[7], so $v^2/c^2 = 2GM/(Rc^2) = r_s/R$.

Schwarzschild gravitational time dilation is the STE (Equation 10) evaluated under the constraint $M_1 = M_2$ (same mass, different radii). Lorentz time dilation is the STE evaluated under the constraint $R_1 = R_2$ (same radius, different masses). Both are cases of the same compactness-based relation; the derivations of each are developed in Section 6.

6 Expressions of Relative Inertial Linear Scale

The universal scale factor between any two systems is the compactness ratio k_d (Equation 10), with $k_d = D_1/D_2 = M_1R_2/(M_2R_1)$. Each subsection below applies a specific physical constraint that simplifies k_d, derives the resulting form, and demonstrates the calculation with a worked numerical example.

Numerical values are reported at six significant figures in e-style notation. Reference constants used throughout: $D_{crit} = 6.73295 \times 10^{26}$ kg/m, $G = 6.67430 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $c = 2.99792 \times 10^8$ m/s. Example parameter sets match the columns of the pair comparison tables (Appendix Appendix A).

6.1 Static Mass (Gravitational) ($M_1 = M_2$)

When the two systems share the same mass, the mass terms cancel in k_d:

$$k_d = \frac{M_1R_2}{M_2R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This case is the standard gravitational time-dilation calculation; the $M_1R_2/(M_2R_1)$ ratio reduces to the familiar Schwarzschild factor here.

Example (Static Mass): S_1: $M_1 = 2.00000 \times 10^{30}$ kg, $R_1 = 3.00000 \times 10^3$ m (near the Schwarzschild radius). S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = 7.00000 \times 10^8$ m.

Compactness ratio:

$$k_d = \frac{R_2}{R_1} = \frac{7.00000 \times 10^8}{3.00000 \times 10^3} = 2.33333 \times 10^5$$

GTD calculation: Schwarzschild gravitational time dilation directly. With $M_1 = M_2$, both systems share the same Schwarzschild radius:

$$r_{s,1} = r_{s,2} = \frac{2GM}{c^2} = \frac{2(6.67430 \times 10^{-11})(2.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$
$$= 2.97046 \times 10^3 \text{ m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\text{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = \sqrt{1 - \frac{2.97046 \times 10^3}{3.00000 \times 10^3}}$$
$$= \sqrt{1 - 9.90155 \times 10^{-1}} = 1 - 9.00777 \times 10^{-1}$$

$$\text{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = \sqrt{1 - \frac{2.97046 \times 10^3}{7.00000 \times 10^8}}$$
$$= \sqrt{1 - 4.24352 \times 10^{-6}} = 1 - 2.12176 \times 10^{-6}$$

giving the relative pace of time:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{1 - 9.00777 \times 10^{-1}}{1 - 2.12176 \times 10^{-6}} = 1 - 9.00776 \times 10^{-1}$$

See Table 2, column $M = M$.

6.2 Static Radius (Linear) ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_d:

$$k_d = \frac{M_1R_2}{M_2R_1} = \frac{M_1}{M_2}$$

The compactness ratio reduces to a ratio of masses. The Lorentz factor (Equation 18) has the same form as Equation 14, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling.

Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius): S_1: $M_1 = 1.00000 \times 10^{30}$ kg, S_2: $M_2 = 1.00000 \times 10^{24}$ kg, both at $R = 1.00000 \times 10^8$ m.

Compactness ratio:

$$k_d = \frac{M_1}{M_2} = \frac{1.00000 \times 10^{30}}{1.00000 \times 10^{24}} = 1.00000 \times 10^6$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^3\,\mathrm{m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{24})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^{-3}\,\mathrm{m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\mathrm{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R}} = \sqrt{1 - \frac{1.48523 \times 10^3}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-5}} = 1 - 7.42619 \times 10^{-6}$$

$$\mathrm{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R}} = \sqrt{1 - \frac{1.48523 \times 10^{-3}}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-11}} = 1 - 7.42616 \times 10^{-12}$$

giving the relative pace of time:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42616 \times 10^{-12}} = 1 - 7.42618 \times 10^{-6}$$

See Table 2, column $R = R$.

6.3 Constant Density (Isometric) ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The compactness ratio reduces to the square of the radius ratio. The linear scale factor follows as $\sqrt{k_d} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density): S_1: $M_1 = 1.00000 \times 10^{30}$ kg, $R_1 = 1.00000 \times 10^8$ m. S_2: $M_2 = 1.00000 \times 10^{27}$ kg, $R_2 = 1.00000 \times 10^7$ m. Both at $\rho = 2.38732 \times 10^5$ kg/m^3.

Compactness ratio and linear scale factor:

$$k_d = \left(\frac{R_1}{R_2}\right)^2 = 1.00000 \times 10^2, \qquad \sqrt{k_d} = \frac{R_1}{R_2} = 1.00000 \times 10^1$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^3\,\mathrm{m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{27})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^0\,\mathrm{m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\mathrm{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = \sqrt{1 - \frac{1.48523 \times 10^3}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-5}} = 1 - 7.42619 \times 10^{-6}$$

$$\mathrm{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = \sqrt{1 - \frac{1.48523 \times 10^0}{1.00000 \times 10^7}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-7}} = 1 - 7.42616 \times 10^{-8}$$

giving the relative pace of time:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42616 \times 10^{-8}} = 1 - 7.35193 \times 10^{-6}$$

See Table 2, column $\rho = \rho$.

6.4 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia): S_1: $M_1 = 1.00000 \times 10^{32}$ kg, $R_1 = 1.00000 \times 10^7$ m. S_2: $M_2 = 1.00000 \times 10^{30}$ kg, $R_2 = 1.00000 \times 10^8$ m. Both have $I = MR^2 = 1.00000 \times 10^{46}$ kg·m^2.

Compactness ratio:

$$k_d = \left(\frac{R_2}{R_1}\right)^3 = \left(\frac{1.00000 \times 10^8}{1.00000 \times 10^7}\right)^3 = 1.00000 \times 10^3$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s =$

$2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{32})}{(2.99792\times10^8)^2}$$

$$= 1.48523\times10^5\,\mathrm{m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{30})}{(2.99792\times10^8)^2}$$

$$= 1.48523\times10^3\,\mathrm{m}$$

GTD for each system via $\sqrt{1-r_s/R}$:

$$\mathrm{GTD}_1 = \sqrt{1-\frac{r_{s,1}}{R_1}} = \sqrt{1-\frac{1.48523\times10^5}{1.00000\times10^7}}$$

$$= \sqrt{1-1.48523\times10^{-2}} = 1-7.45394\times10^{-3}$$

$$\mathrm{GTD}_2 = \sqrt{1-\frac{r_{s,2}}{R_2}} = \sqrt{1-\frac{1.48523\times10^3}{1.00000\times10^8}}$$

$$= \sqrt{1-1.48523\times10^{-5}} = 1-7.42619\times10^{-6}$$

giving the relative pace of time:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{1-7.45394\times10^{-3}}{1-7.42619\times10^{-6}} = 1-7.44657\times10^{-3}$$

See Table 2, column $I = I$.

6.5 General Case (unconstrained)

When no constraint is applied between the systems, k_d stays in its full form:

$$k_d = \frac{M_1R_2}{M_2R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case): S_1: $M_1 = 1.00000\times10^{20}$ kg, $R_1 = 2.00000\times10^{-7}$ m. S_2: $M_2 = 1.00000\times10^{30}$ kg, $R_2 = 1.00000\times10^7$ m.

Compactness ratio:

$$k_d = \frac{M_1R_2}{M_2R_1} = \frac{(1.00000\times10^{20})(1.00000\times10^7)}{(1.00000\times10^{30})(2.00000\times10^{-7})} = 5.00000\times10^3$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{20})}{(2.99792\times10^8)^2}$$

$$= 1.48523\times10^{-7}\,\mathrm{m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{30})}{(2.99792\times10^8)^2}$$

$$= 1.48523\times10^3\,\mathrm{m}$$

GTD for each system via $\sqrt{1-r_s/R}$:

$$\mathrm{GTD}_1 = \sqrt{1-\frac{r_{s,1}}{R_1}} = \sqrt{1-\frac{1.48523\times10^{-7}}{2.00000\times10^{-7}}}$$

$$= \sqrt{1-7.42617\times10^{-1}} = 1-4.92670\times10^{-1}$$

$$\mathrm{GTD}_2 = \sqrt{1-\frac{r_{s,2}}{R_2}} = \sqrt{1-\frac{1.48523\times10^3}{1.00000\times10^7}}$$

$$= \sqrt{1-1.48523\times10^{-4}} = 1-7.42644\times10^{-5}$$

giving the relative pace of time:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{1-4.92670\times10^{-1}}{1-7.42644\times10^{-5}} = 1-4.92632\times10^{-1}$$

See Table 2, column General.

6.6 Inertia–Time Dilation Identities

The preceding cases derive time dilation from inertia. The relationship is invertible: given the compactness ratio and the radius ratio, the moment of inertia ratio is fully determined. From Equation 10:

$$\frac{I_1}{I_2} = k_d\cdot\left(\frac{R_1}{R_2}\right)^3 \tag{19}$$

The inertia ratio decomposes into a compactness ratio and a spatial scale factor (the volume ratio). This is the STE (Equation 10) inverted: the $(R_1/R_2)^3$ factor converts the R^{-1} dependence in DeGerlia compactness back to the R^2 dependence in inertia.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste`.

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Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 6: static mass (M=M), static radius (R=R), constant density ($\rho = \rho$), constant inertia (I=I), an unconstrained general case, and an STE example. The isometric equivalence scale factor $(I_1/I_2)^{1/5}$ equals $\sqrt{k_d}$ only under constant density (Case 3 and the STE Example); in all other cases the universal STE (Equation 16) with its density correction is required. The inertia recovery identity $k_d(R_1/R_2)^3 = I_1/I_2$ holds in every case. The table is thus the numerical demonstration both of the original fifth-root inertia hypothesis (exact in the isometric column) and of its corrected universal form (exact in every column).

Table 1 System properties for each pair comparison case.

	Case 1 ($M = M$)	Case 2 ($R = R$)	Case 3 ($\rho = \rho$)	Case 4 ($I = I$)	General	STE Example
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
m_2 (kg)	2.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
r_1 (m)	3.00000e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	1.76839e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	1.39203e+3	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	7.20000e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	3.92000e+47	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.66667e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
D_{1_norm}	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
D_{2_norm}	4.24352e-6	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
r_{s_1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s_2} (m)	2.97046e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
gtd_1	1-9.00777e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
gtd_2	1-2.12176e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28

Table 2 Derived ratios across constraint cases.

	Case 1 ($M = M$)	Case 2 ($R = R$)	Case 3 ($\rho = \rho$)	Case 4 ($I = I$)	General	STE Example
$\sqrt{D_1/D_2}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(m_1 r_2)/(m_2 r_1)$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(r_2/r_1)^3$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
gtd_1/gtd_2	1-9.00776e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
I_1/I_2	1.83673e-11	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.28571e-6	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
m_1/m_2	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
r_1/r_2	4.28571e-6	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{r_1/r_2}$	2.07020e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0